Muhammad Irfan

Machine Learning Engineer

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D-2-3(Master's Student Visa)

EXPERIENCE

Research student 09/2025 - Present
Dong-A University ↗

Working as a graduate research student in Data Science with a focus on medical image analysis.

- Developing machine learning models for image classification tasks.
- Applying visualization techniques such as Grad-CAM to interpret model predictions.
- Analyzing and improving model performance using real-world medical datasets.

Self-Directed Data Science Project 03/2026 - 03/2026
Insurance Charges Prediction (Machine Learning Project) ↗

- I built a Linear Regression model to predict medical insurance charges based on patient features.
- I achieved a model performance of **80.4% R^2 score** and **79.9% Adjusted R^2 score** on the test dataset.
- I performed data preprocessing including encoding categorical variables and feature scaling.
- I conducted exploratory data analysis (EDA) to understand relationships between age, BMI, smoking status, and region.

Self-Learning Project in Machine Learning 02/2026 - 02/2026
Cardio-AI: Heart Disease Prediction System ↗

- I used multiple machine learning models for heart disease prediction.
- I achieved the best performance using the KNN model with **88.6% accuracy** and **89.8% F1-score.**
- I compared Logistic Regression, SVM, and Naive Bayes models for performance evaluation.
- I developed a Streamlit web application for real-time prediction with probability outputs.

EDUCATION

Data Science (Medical Image Analysis) 09/2025 - Present
Dong-A University - Data Science Convergence

- Focus on **Medical Image Analysis.**
- Relevant Coursework: Basic Mathematics, Medical Statistics, Machine Learning.
- Developing skills in applying data science techniques to healthcare data.

Mathematics ↗ 09/2019 - 09/2023
University of Okara - Bachelor of Science

- Studied core subjects including Numerical Analysis, Linear Algebra, Calculus, and Statistics.
- Gained strong analytical and problem-solving skills through mathematical training.
- Learned Object-Oriented Programming and basic computational techniques.

CERTIFICATES

Intermediate Data Science with Machine learning 10/2024 - 01/2025
Knowledge Streams

- Gained practical experience in Data Science and Machine Learning through hands-on projects.

- Solved coding challenges and applied concepts to real-world problems.
- Collaborated in study groups and workshops to enhance learning and teamwork.

Data Visualization with Python ↗ 10/2023 - 12/2023
IBM

- Created visualizations using Matplotlib, Seaborn.
- Built charts such as line, bar, histogram, pie, and scatter plots.
- Developed advanced visualizations including maps and dashboards.
- Used Dash framework to build interactive data dashboards.

Data Analysis with Python ↗ 10/2023 - 11/2023
IBM

- Learned data cleaning, preprocessing, and handling missing values.
- Performed Exploratory Data Analysis (EDA) using Pandas, NumPy.
- Applied data visualization techniques using Matplotlib.
- Built and evaluated regression models using Scikit-learn.

SKILLS

FastAPI	Intermediate	Deep learning	Intermediate
Tensor flow	Intermediate	Machine Learning	Intermediate
Microsoft Excel	Advanced	Python	Intermediate

LANGUAGES

Korean	Basic	English	Fluent